

Q1

Data:-

$$C = 1.3 \text{ PF} = 1.3 \times 10^{-12} \text{ F}$$

$$C' = 2.6 \text{ PF} = 2.6 \times 10^{-12} \text{ F}$$

$$\epsilon_r = ?$$

Sol:-

for Air

$$C = \frac{\epsilon_0 A}{d} \rightarrow (i)$$

for dielectric

$$C' = \frac{\epsilon_0 \epsilon_r A}{d}$$

$\therefore$ condition

$$C' = \frac{\epsilon_0 \epsilon_r d A}{2d} \rightarrow (ii')$$

Divide (2) by (1)

$$\frac{C'}{C} = \frac{\epsilon_0 A \epsilon_r / 2d}{\epsilon_0 A / d} = \frac{\epsilon_0 \epsilon_r A}{2d} \times \frac{d}{\epsilon_0 A}$$

$$\frac{C'}{C} = \frac{\epsilon_r}{2}$$

$$\therefore \epsilon_r = \frac{2C'}{C} = \frac{2(2.6 \times 10^{-12})}{1.3 \times 10^{-12}}$$

$$\boxed{\epsilon_r = 4}$$

Q2

Data:-

$$C_1 = 10 \mu F$$

$$C_2 = 50 \mu F$$

$$C_3 = 25 \mu F$$

$$V = 250V$$

i) $q_1 = ?$	ii) $C_{eq} = ?$
$q_2 = ?$	
$q_3 = ?$	

connected in parallel

Sol:-

i)

$$q = CV$$

$$q_2 = CV$$

$$q_1 = C_1 V$$

$$q_2 = (50 \times 10^{-6})(250)$$

$$q_1 = (10 \times 10^{-6})(250)$$

$$q_2 = 12.5 \text{ mC}$$

$$q_1 = 2.5 \text{ mC}$$

$$q_3 = C_3 V$$

$$q_3 = (25 \times 10^{-6})(250)$$

$$q_3 = 6.25 \text{ mC}$$

ii)

$$C_{eq} = C_1 + C_2 + C_3$$

$$C_{eq} = 10 + 50 + 25$$

$$C_{eq} = 85 \mu F$$

iii)

$$q = q_1 = q_2 = q_3$$

$$\therefore \frac{1}{C_{Eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$$

$$\frac{1}{C_{Eq}} = \frac{1}{10} + \frac{1}{50} + \frac{1}{25}$$

$$\frac{1}{C_{Eq}} = \frac{5 + 1 + 2}{50} = \frac{8}{50}$$

$$\Rightarrow \frac{1}{C_{Eq}} = \frac{8}{50}$$

$$\Rightarrow C_{Eq} = \frac{50}{8} = 6.25 \mu F$$

$$\therefore q = CV$$

$$q = (6.25 \times 10^{-6})(250)$$

$$q = 1.5625 \times 10^{-3} C = q_1 = q_2 = q_3$$

$$\therefore V_1 = \frac{q}{C_1} = \frac{1.562 \times 10^{-3}}{10 \times 10^{-6}}$$

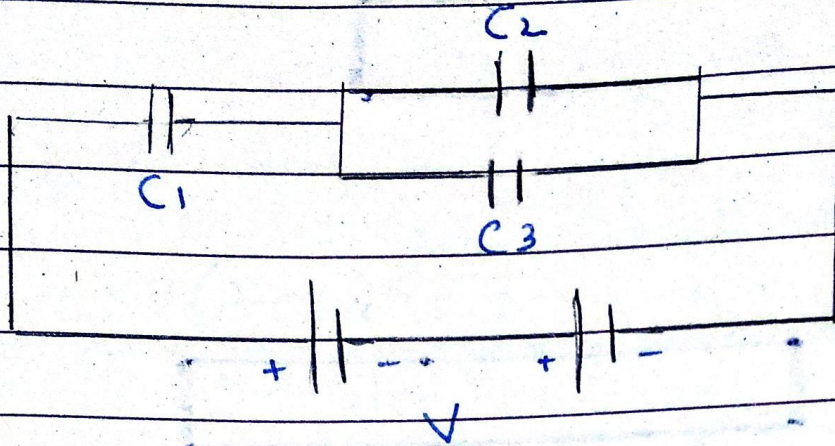
$$V_1 = 156.25 V$$

$$V_2 = \frac{q}{C_2} = \frac{1.5625 \times 10^{-3}}{50 \times 10^{-6}} = 31.25 V$$

$$V_3 = \frac{q}{C_3} = \frac{1.5625 \times 10^{-3}}{25 \times 10^{-6}}$$

$$V_3 = 62.5 \text{ V}$$

Q3



Data:-

$$V = 100 \text{ V}$$

$$C_1 = 2 \mu\text{F}$$

$$C_2 = 3 \mu\text{F}$$

$$C_3 = 5 \mu\text{F}$$

$$q_1 = ?, q_2 = ?, q_3 = ?$$

$$V_1 = ?, V_2 = ?, V_3 = ?$$

Sol:-

C_2 and C_3 are connected in parallel. Then their Equivalent C' is given as:

$$C' = C_2 + C_3$$

$$C' = 3 + 5$$

$$C' = 8 \mu\text{F}$$

Now C_1 and C' are connected in series:

$$\therefore C_{Eq} = \frac{C_1 \times C'}{C_1 + C'} = \frac{2 \times 8}{2 + 8} =$$

$$C_{Eq} = 1.6 \mu F$$

$$\therefore q_{Eq} = C_{Eq} V$$

$$q_{Eq} = (1.6 \times 10^{-6}) \times 100$$

$$q_{Eq} = 1.6 \times 10^{-4} C$$

Now C_1 and C' are in series both have same charge
for C_1 ,

$$\therefore q_1 = q' = q_{Eq} = 1.6 \times 10^{-4} C$$

$$q_1 = 1.6 \times 10^{-4} C \quad (\text{or}) \quad 160 \mu C$$

$$V_1 = \frac{Q_1}{C_1} = \frac{1.6 \times 10^{-4}}{2 \times 10^{-6}} = 80V$$

$$V_1 = 80V$$

for C' ,

Voltage across C' ,

$$V' = \frac{Q_1}{C'}$$

$$V' = \frac{1.6 \times 10^{-4}}{8 \times 10^{-6}} = 20V$$

Date _____

$\therefore$ Voltage across C_2 and C_3 is 20V

$$\therefore Q = CV$$

$$Q_2 = C_2 V_2 \quad ; \quad Q_3 = C_3 V_2$$

$$Q_2 = (3 \times 10^{-6})(20) \quad ; \quad Q_3 = (5 \times 10^{-6})(20)$$

$$Q_2 = 6 \times 10^{-5} \quad ; \quad Q_3 = 100 \mu C$$

$$Q_2 = 60 \mu C$$

Answers: $Q_1 = 160 \mu C$, $Q_2 = 60 \mu C$, $Q_3 = 100 \mu C$
 $V_1 = 80V$, $V_2 = 20V$, $V_3 = 20V$

Q4

Data

$$\text{Capacitance} = C = 10 \mu\text{F} = 10 \times 10^{-6} \text{ F}$$

$$\text{Resistance} = R = 1 \text{ M}\Omega = 1 \times 10^6 \Omega$$

$$Q = 90\% Q_0 = 0.9 Q_0$$

$$\text{time taken} = t = ?$$

Sol:-

for charging of capacitor

$$Q = Q_0 [1 - e^{-t/RC}]$$

$$0.9 Q_0 = Q_0 [1 - e^{-t/RC}] \rightarrow (i)$$

$$RC = 1 \times 10^6 \times 10 \times 10^{-6}$$

$$RC = 10$$

$$0.9 = 1 - e^{-t/10}$$

$$e^{-t/10} = 1 - 0.9$$

taking natural log o.b.s

$$\ln e^{-t/10} = \ln(0.1)$$

$$-\frac{t}{10} = -2.302$$

$$t = 23 \text{ sec}$$

Q35

Data:-

$$\text{Energy} = U = 10 \text{ kWh} = 10 \times 1000 \times 3600$$

$$U = 3.6 \times 10^7 \text{ J}$$

$$V = 1000 \text{ V}$$

$$C = ?$$

Sol

$$U = \frac{1}{2} CV^2$$

$$C = \frac{2(3.6 \times 10^7)}{(1000)^2}$$

$$C = \frac{2U}{V^2}$$

$$C = 72 \text{ Farad}$$

Q6

Data: $C_1 = 2\mu$

$C_2 = 4\mu F$

$V = 300V$

total energy = $U = ?$

Sol:-

Since C_1 and C_2 are in parallel

$$C_{eq} = C_1 + C_2$$

$$C_{eq} = 6\mu F$$

$$U = \frac{1}{2} CV^2$$

$$U = \frac{1}{2} (6 \times 10^{-6}) (300)^2$$

$$U = 0.27J$$

Q7

Data:-

$$V = 12V$$

$$Q = 54\mu C = 54 \times 10^{-6} C$$

$E = u = ?$

Sol:-

$$u = \frac{1}{2} QV$$

$$U = \frac{1}{2} (54 \times 10^{-6}) (12)$$

Date _____

$$U = 3.24 \times 10^{-4} \text{ J}$$